

Exp No: 9

ULTRASONIC SENSOR IN PROTEUS USING ARDUINO

OBJECTIVES

- To interface an Ultrasonic Sensor with Arduino Uno using Proteus simulation.
- To measure the distance of an object using ultrasonic waves.
- To display the measured distance through the Virtual Terminal in Proteus.

PROGRAM

// Program Sketch

```
int trig = 10;
```

```
int echo = 9;
```

```
long duration;
```

```
int cm;
```

```
void setup() {
```

```
// put your setup code here, to run once:
```

```
pinMode(trig, OUTPUT);
```

```
pinMode(echo, INPUT);
```

```
Serial.begin(9600);
```

```
}
```

```
void loop() { // Sketch to calculate ultrasonic distance
```

```
digitalWrite(trig, LOW);
```

```
delayMicroseconds(10);
```

```
digitalWrite(trig, HIGH);
```

```
delayMicroseconds(10);
```

```
digitalWrite(trig, LOW);
```

```
delayMicroseconds(10);
```

```
duration = pulseIn(echo, HIGH);
```

```
cm = (duration/2) * 0.034;
```

}

